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RESEARCH ARTICLE

Rapid scavenging of avian carcasses in the desert Southwest: Implications for mortality surveys not associated with infrastructure

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Abstract

Avian populations have experienced widespread declines with mortality attributable to a range of factors. Carcass surveys have been employed to document and quantify avian mortality in both infrastructure-related and non-infrastructure-related scenarios. To improve the accuracy of mortality estimates obtained from carcass surveys, correction factors can be used to control for sources of bias such as the removal of carcasses by scavengers. However, given differences in how carcasses appear on the landscape between infrastructure-related mortality and non-infrastructure-related mortality events it is possible that estimates of carcass persistence from infrastructure-focused studies may not be relevant for all mortality monitoring. We conducted a study of carcass persistence on White Sands Missile Range in the Tularosa Basin of New Mexico, USA, simulating mortality unrelated to infrastructure. We placed and monitored 214 avian carcasses over 2 seasons, determined how long each carcass persisted prior to being scavenged, and examined the influence of habitat, temporal, and scavenger guild variables using Weibull survival models. We found that 211 of the 214 placed carcasses were scavenged and median persistence time was 1.53 days. Our analysis showed that the null model was the most competitive, and the 85% confidence intervals for all predictor variables

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overlapped zero, indicating none were informative for carcass persistence times. Overall, our findings suggested that persistence times in a non-infrastructure-related mortality scenario are similar to those associated with infrastructure, underscoring the need to account for rapid scavenger removal when designing mortality monitoring programs.

KEYWORDS

avian conservation, avian mortality, carcass persistence, carcass scavenging, carcass surveys, scavenger removal, scavengers

Avian populations have experienced widespread declines as the result of direct and indirect impacts of anthropogenic activities (Stanton et al. 2018, Rosenberg et al. 2019, Katzner et al. 2020, Rigal et al. 2023). Direct mortality of birds from anthropogenic causes is well documented, particularly due to collisions with buildings (Loss et al. 2014a), vehicles (Loss et al. 2014b), power lines (Bernardino et al. 2018), and wind turbines (Loss et al. 2013, 2015). Accurate estimates of mortality are essential for quantifying the impact of specific threats on bird populations, guiding targeted mitigation and remediation efforts, and ensuring permitting compliance for infrastructure-related mortality (Ravache et al. 2024). Given the importance of mortality estimates and the legal ramifications related to permitting, informed by the estimates, methods and best practices to obtain accurate mortality estimates have received a great amount of research attention.

Mortality surveys, which typically involve teams of humans or conservation detection dogs systematically searching designated areas for avian carcasses (Smallwood 2007, Santos et al. 2011, Rioux et al. 2013, Barrientos et al. 2018, Boland et al. 2026), are widely used to quantify the lethal impacts of anthropogenic activities and provide information for assessment of population-level impacts to birds. Although surveys are commonly used to estimate mortality, mortality surveys have inherent biases that must be corrected to estimate true mortality (Kunz et al. 2007, Smallwood 2007, Barrientos et al. 2018). One primary source of bias in mortality surveys is the removal of carcasses by scavengers before they can be detected by surveyors, which can lead to underestimation of true mortality rates (Borner et al. 2017). To address removal bias, it is necessary to account for carcasses that may have been removed by scavengers prior to survey efforts. Previous studies have identified several factors contributing to the variability in carcass persistence times related to scavenger removal of carcasses, including time of year, surrounding habitat, carcass size, and scavenger species (Flint et al. 2010, Bispo et al. 2013, Barrientos et al. 2018, Varela and Zimmerman 2019, Ravache et al. 2024). Additionally, the spatial distribution of carcasses on the landscape, and the predictability of where carcasses may occur, has been linked to scavenger behavior and variability in persistence times (Santos et al. 2011, Fielding et al. 2021).

While studies have quantified carcass persistence times for mortality associated with vertical infrastructure like solar arrays, buildings, wind turbines, and powerlines (Barrientos et al. 2018), much less research attention has focused on mortality scenarios not associated with vertical infrastructure (hereafter, infrastructure), such as mass avian mortality events associated with extreme weather. Differences in carcass distribution between infrastructure and non-infrastructure scenarios may drive variability in carcass persistence times. In infrastructure-related scenarios, carcasses tend to cluster in predictable locations and appear regularly, allowing scavengers to habituate to a search area, potentially leading to shortened persistence times (Santos et al. 2011, Fielding et al. 2021). In contrast, mass avian mortality events caused by extreme weather vary widely in the number of carcasses produced, the area over which carcasses occur, and the density of carcasses on the landscape (Newton 2007). In non-infrastructure-related scenarios, carcasses may be dispersed unpredictably (in time and space) across the landscape, and over a greater area, preventing scavengers from habituating to carcass presence in specific areas. Additionally, carcasses

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may occur at high densities, which may swamp scavengers, or at low densities, which may allow for complete removal. As a result, carcass persistence times derived from infrastructure-related mortality studies may not be directly comparable to carcass persistence times in non-infrastructure-related scenarios.

In 2020, a mass avian mortality event occurred across the American Southwest that resulted in the deaths of thousands of birds throughout the region (Yang et al. 2021, Boland 2024). Carcasses were found both incidentally and through survey efforts over several weeks across diverse habitats and not always associated with infrastructure, with mortality primarily resulting from starvation associated with widespread drought and extreme weather, even for carcasses found near infrastructure. With climate change projected to intensify in the region (Elias et al. 2023), similar avian mortality events may recur, highlighting the need to accurately quantify their impacts on bird populations.

Given that mass mortality events unrelated to infrastructure may continue to occur, it is important to understand how avian carcass persistence times differ between infrastructure- and non-infrastructure-related mortality scenarios. Our objectives were to quantify avian carcass persistence times in an arid landscape and to evaluate the effects of seasonality, vegetation cover, and proximity to roads on carcass persistence and scavenging rates associated with non-infrastructure related mortality. To address our objectives, we conducted a study of avian carcass scavenging in southern New Mexico, USA, during the spring and fall of 2022. We randomly placed carcasses across the landscape, typically not in association with infrastructure, matching the locations where carcasses appeared during the 2020 avian mortality event in the region. Studies of avian carcass persistence associated with infrastructure in arid lands have found average persistence times of approximately 2–3 days (Barrientos et al. 2018). We predicted that persistence times of carcasses in our system, which were not associated with infrastructure, would be longer than those associated with infrastructure as scavengers would not be able to habituate to carcass presence. We predicted that persistence times would be shorter in the fall than in the spring due to increased scavenger abundance following the breeding season, and as previous studies have found longer persistence times in the dry season (spring in our study region) than in the wet season (Flint et al. 2010, Hager et al. 2012, Villegas-Patraca et al. 2012). Additionally, due to relationships with carcass detectability, we predicted that carcasses would persist longer in areas with greater vegetation cover and visual obstruction (Hager et al. 2012, Barrientos et al. 2018, Varela and Zimmerman 2019). Also, we predicted that carcasses placed closer to sunrise would persist longer than carcasses placed later in the day, as we expected most scavengers to be nocturnal in our system. Lastly, we predicted distance to roadways would be positively related to carcass persistence times due to scavenger behaviors in relation to predictable carrion occurring on roadways (Santos et al. 2011, Fielding et al. 2021).

STUDY AREA

Our study was conducted on White Sands Missile Range, which is located within the Tularosa Basin in south-central New Mexico (32°26′42.92″N, 106°19′1.92″W), USA (Figure 1). The Tularosa Basin, in the northern portion of the Chihuahuan Desert ecoregion, is surrounded by the Organ, Franklin, Sacramento, and San Andres Mountains and is a desert-scrub environment dominated by honey mesquite (*Prosopis glandulosa*) and creosote bush (*Larrea tridentata*). The area experiences mild winters and hot summers with seasonal monsoons from July to September, a weather regime similar to much of the American Southwest. Average monthly high temperature within the region ranges from 13°C in December to 35°C in June, while average monthly low temperature ranges from −2°C in January to 19°C in July (US Climate Data 2024). Average annual precipitation in the Tularosa Basin is 33 cm per year, with most precipitation falling during the monsoon season (US Climate Data 2024). In comparison to surrounding urban areas, human infrastructure is relatively sparse on White Sands Missile Range. The density of buildings is low throughout most of the Missile Range, and while roadways do exist, most see little traffic.

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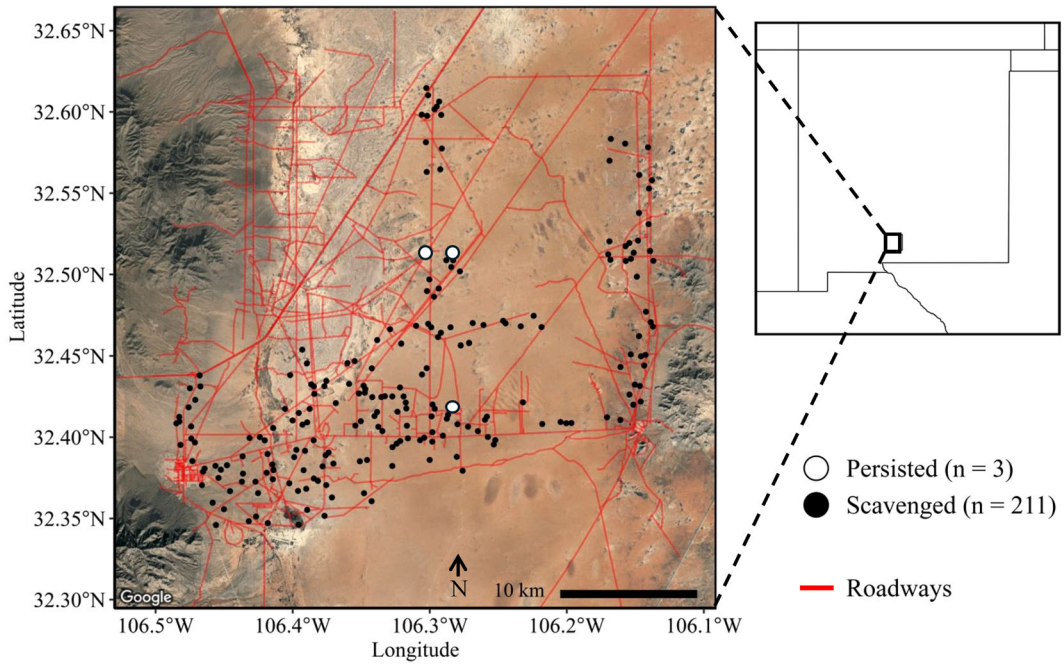


FIGURE 1 Map of the study area, roadways, and avian carcass placement locations ($n = 214$) for all carcasses placed not in association with infrastructure on White Sands Missile Range in south-central New Mexico, USA, during spring (April–May) and fall (September–November) 2022.

METHODS

Following the recommendations of Barrientos et al. (2018), we chose to use house sparrow (*Passer domesticus*) carcasses for our persistence trials, as they resemble the typical size and shape of avian migrants that pass through the study area and were readily accessible. House sparrow carcasses were collected by capturing and euthanizing house sparrows at a location outside of the study area, using methods approved by New Mexico State University's Institutional Animal Care and Use Committee (protocol No. 2022-028). Captured house sparrows were euthanized using rapid cardiac compression to prevent residual odors from chemical euthanasia and to avoid scavengers ingesting chemical agents. Carcasses were frozen immediately following euthanasia and handled exclusively with gloved hands.

To align with peak periods of avian activity within our region, which correspond to seasonal migration and when previous mortality events have occurred (Boland 2024, Osterhaus et al. 2025), we conducted carcass persistence trials during the spring (1 April to 31 May) and fall (1 September to 10 November) of 2022. To match the random distribution of carcasses on the landscape during the 2020 mass avian mortality event, we generated a set of random points using ArcGIS to serve as locations for carcass placements on White Sands Missile Range (Figure 1). Random points were constrained by the following requirements: 1) a minimum distance of 200 m between points to preclude scavenger swamping and representative of carcass arrangement in previous mortality events, and 2) points must be within 750 m of a road to ensure accessibility, but no closer than 10 m of a roadway to reduce the risk of collisions with vehicles. Carcasses were removed from freezer storage and allowed to thaw at room temperature for 12 hours prior to placement (Barrientos et al. 2018). Carcasses were placed during daylight hours with a maximum of 15 carcasses present on the landscape at any one time to prevent scavenger swamping and allow for placements to occur within a single day. Placement locations were used only once during

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the study period. Carcasses were placed as near to the random point as possible; however, in instances where dense woody vegetation (typically mesquite hummocks) obscured the immediate vicinity of a designated point, carcasses were placed at the nearest edge of such vegetation, enabling visibility for the camera traps.

Each carcass was monitored with a Reconyx HC600 camera trap (Reconyx, Holmen, WI, USA) mounted onto a wooden stake at a height of ~1 m and positioned ~3 m from the carcass (Paula et al. 2014, Rosa et al. 2019). All camera traps faced northwards to minimize sun glare. Cameras were set to the highest sensitivity and programmed to capture a sequence of 5 images (one every 3 seconds) following activation by motion. In addition to the motion trigger, cameras were also set to automatically capture a single image every thirty minutes throughout the monitoring period, as previous studies have noted low detection rates of scavenging events (Paula et al. 2014). If no image directly captured the scavenger, photos enabled us to quantify carcass persistence time with a maximum error of 30 minutes. All placement locations were revisited after one week to evaluate carcass persistence. Carcasses were deemed scavenged if no remains identifiable to species were found within a 10-m radius of their original placement, adhering to criteria established by Paula et al. (2014). If the carcass persisted after the first week, it was monitored for an additional week. If the carcass was still present after 2 weeks, the carcass was classified as persisted, removed, and monitoring of that placement location was concluded.

A manual review of photographs was conducted to determine the persistence time of each carcass and to identify and classify scavenger species and guild. Scavengers were classified into the following guilds: small mammal (rodents), medium mammal (mesocarnivores), large mammal, avian, invertebrate, or reptile. In cases where scavengers could not be identified at the species level, identifications were made at the lowest taxonomic level possible. For instance, gray fox (*Urocyon cinereoargenteus*) and kit fox (*Vulpes macrotis*) were undifferentiated in many photographs, therefore we grouped these 2 species as Fox spp.

For every site at which a carcass was placed, the surrounding vegetation structure (cover and visual obstruction) was quantified. For cover, a point-intercept transect-based methodology was employed (Godínez-Alvarez et al. 2009, Bonham 2013). At each site, 4 10-m transects were established, extending from the carcass placement location in each cardinal direction. Along each transect, percent vegetation cover and visual obstruction were measured using the line-point intercept and Robel pole methods, respectively (Robel et al. 1970). Distance to the nearest road was calculated for each placement location using Google Earth Pro. Placement time was defined as the time in hours between sunrise and the time at which the carcass was placed. Day placed was defined as the ordinal day on which the carcass was placed. Season corresponded to the season in which the carcass was placed (either spring or fall).

Data analysis

We fitted a suite of parametric survival models using the survreg() and Surv() functions from the survival package in R, with type set to interval, to assess how our suite of covariates (scavenger guild, placement time, season, percent vegetation cover, visual obstruction, and distance to road) may influence carcass persistence times (R version 4.5.0; Therneau et al. 2024, R Core Team 2025). For carcasses where the exact moment of scavenging was not captured by the camera traps, we used the 30 min timelapse with time equal to the time of the last photograph where the carcass was present and time2 equal to the time of the first photograph where the carcass was absent. Prior to model fitting, we assessed collinearity between continuous predictor variables (placement time, percent vegetation cover, visual obstruction, and distance to road) using the Pearson correlation coefficient, and any highly correlated variables ($|r| > 0.7$) were not included within the same model. All continuous covariates were scaled and centered prior to modeling. To identify the most appropriate distribution for subsequent carcass persistence analyses, we fitted null models for 4 candidate distributions (Weibull, exponential, log-normal, log-logistic) and compared them using the Akaike Information Criterion adjusted for small sample sizes (AIC_c; Burnham and Anderson 2002).

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The Weibull (AIC_c = 2269.47) and exponential (AIC_c = 2267.59) distributions were competitive (i.e., within 2 AIC_c of the top model); however, we chose the Weibull distribution for our subsequent analyses due to its flexibility.

Our suite of models included null, global, and univariate models for each predictor. We also defined a set of more complex models by grouping our predictor variables into 3 categories, and creating additive models based on the combinations of these categories. The 3 categories were: Habitat (percent vegetation cover, visual obstruction, and distance to road), Temporal (season [spring or fall] and placement time), and Guild (guild).

We conducted model selection for the suite of models using AIC_c. Models within 2 AIC_c of the top model were considered to have comparable support and were identified as competitive models. Covariates were considered important predictors of carcass persistence time if the 85% confidence interval for their estimated coefficient did not overlap 0. We used 85% confidence intervals, which is consistent with AIC-based model selection strategy (Arnold 2010, Sutherland et al. 2023). For categorical predictors, the reference category for season was spring, and the reference category for guild was avian. All figures were generated using the ggplot2 package in R (Wickham et al. 2024).

RESULTS

Of the 214 carcasses placed, 211 were scavenged, while only 3 persisted for the full 2-week monitoring period (Figure 2). For the 211 scavenged carcasses, persistence time ranged from 17 minutes to 11.48 days. Photographic evidence was available for 104 of the 211 scavenged carcasses, allowing for both identification of the scavenger and exact determination of persistence time (Figures 2, 3). For the remaining 107 scavenged carcasses, no photographic evidence of the scavenger was available. Of scavenging events where the scavenger could be identified, the medium mammal guild was responsible for the most events, and we identified a minimum of 11 scavenger species (Figure 2). At the species level, the Mexican (MX) ground squirrel (*Spermophilus mexicanus*) accounted for

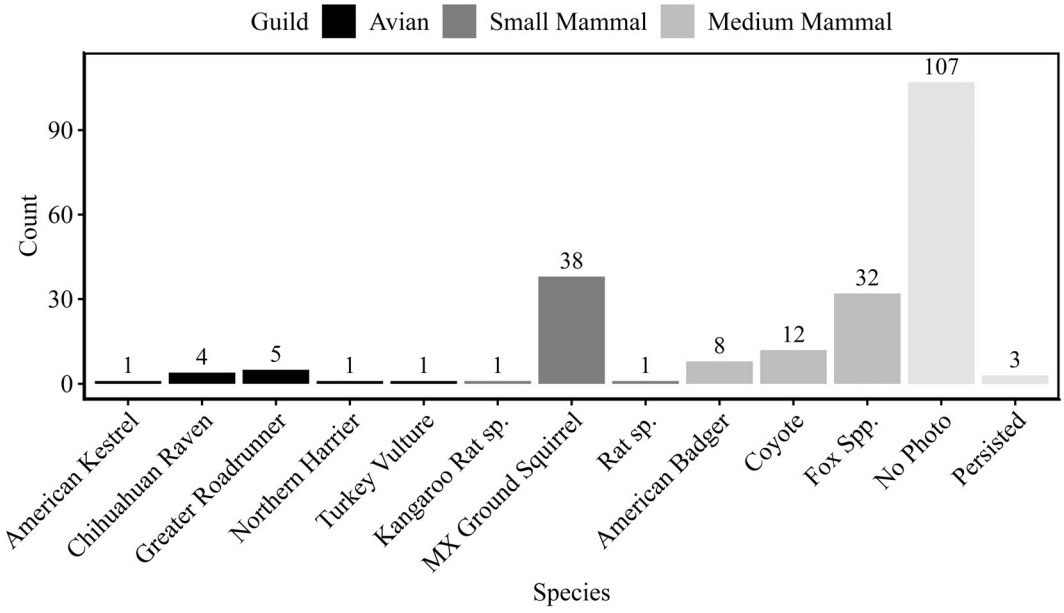


FIGURE 2 Summary of fates of carcass placements for the 214 carcasses placed at random locations on White Sands Missile Range in south-central New Mexico, USA, during spring (April–May) and fall (September–November) 2022 and monitored for up to 2 weeks.

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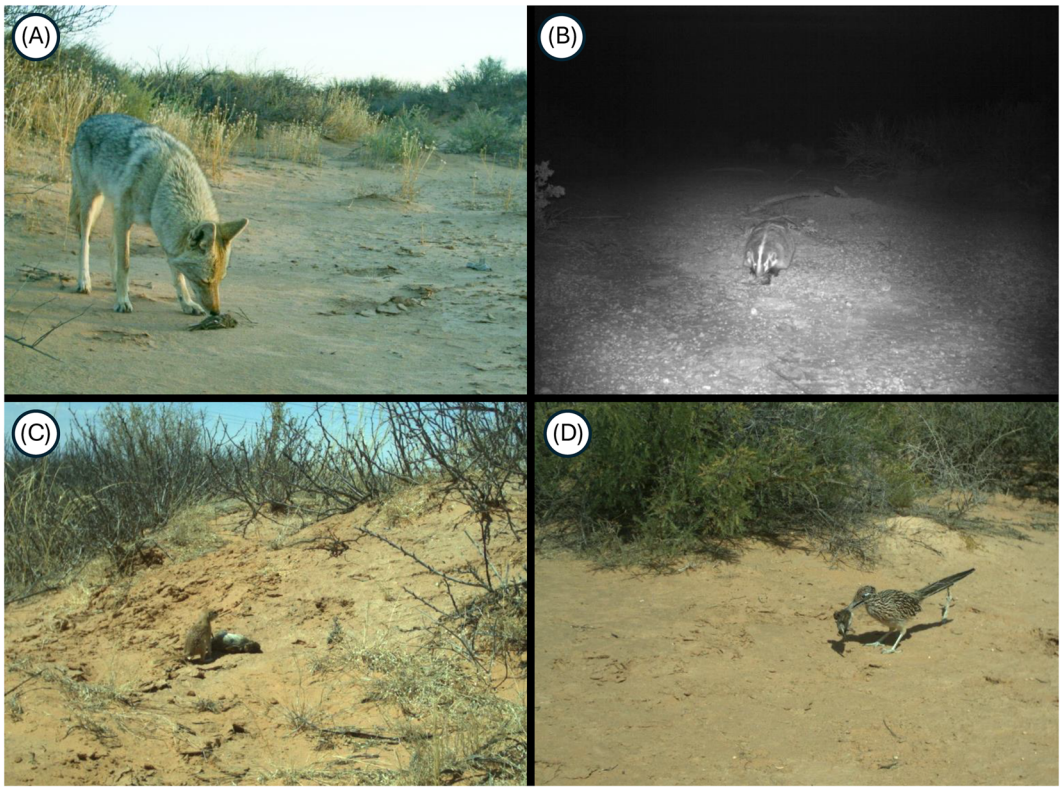


FIGURE 3 Scavengers detected by camera traps during avian carcass persistence trials for carcasses not placed in association with infrastructure on White Sands Missile Range in south-central New Mexico, USA, during spring (April–May) and fall (September–November) 2022 included (A) coyote (*Canis latrans*), (B) American badger (*Taxidea taxus*), (C) Mexican ground squirrel (*Ictidomys mexicanus*), and (D) greater roadrunner (*Geococcyx californianus*).

the highest number of scavenging events, followed by Fox spp. (Figures 2, 3). Median persistence times for carcasses scavenged by avian, small mammal, and medium mammal guilds were 0.92, 1.91, and 2.32 days, respectively. No removal events were recorded for invertebrate, reptilian, or large mammal scavengers. Carcasses without a photo of the scavenging event had shorter median persistence times (1.39 days) than those with a photo (1.89 days).

The null model was the top-ranked model, with univariate models for guild, percent vegetation cover, placement time, and season also considered competitive (within 2 AIC_c units; Table 1). The top ranking of the null model suggests that the tested predictor variables were not informative, indicating that our predictor variables explained little of the variation in carcass persistence times. Therefore, we used the null model to derive the predicted probability of persistence for a carcass over the 2-week monitoring period (Figure 4). Our model predicted that median carcass persistence time was 1.53 days and that the probability of a carcass persisting for 24 hours, 2 days, and 1 week was 0.65, 0.40, and 0.03, respectively.

DISCUSSION

Mortality surveys are a key tool for quantifying avian mortality and assessing population-level threats; however, the accuracy of mortality surveys depends on identifying, quantifying, and correcting for biases associated with carcass detectability. We investigated the impacts of scavenging and factors that may influence carcass persistence times in

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TABLE 1 Model selection results for Weibull survival models of avian carcass persistence time for carcasses placed not in association with infrastructure on White Sands Missile Range in south-central New Mexico, USA, during spring (April–May) and fall (September–November) 2022.

Model Name	Model Formula	# of Parameters	AIC_c	Delta AIC_c
Null	1	1	2230.33	0
Guild	Guild	4	2231.06	0.73
Cover	Cover	2	2231.89	1.56
Time Placed	Placement Time	2	2232.19	1.86
Season	Season	2	2232.24	1.91
Obstruction	Obstruction	2	2232.34	2.01
Distance to Road	Distance to Road	2	2232.35	2.02
Temporal	Season + Placement Time	3	2234.05	3.72
Temporal and Guild	Guild + Season + Placement Time	6	2234.8	4.47
Habitat	Cover + Obstruction + Distance to Road	4	2235.84	5.51
Habitat and Guild	Cover + Obstruction + Distance to Road + Guild	7	2236.32	5.99
Habitat and Temporal	Cover + Obstruction + Distance to Road + Season + Placement Time	6	2239.69	9.36
Global	Cover + Obstruction + Distance to Road + Guild + Season + Placement Time	9	2240.27	9.94

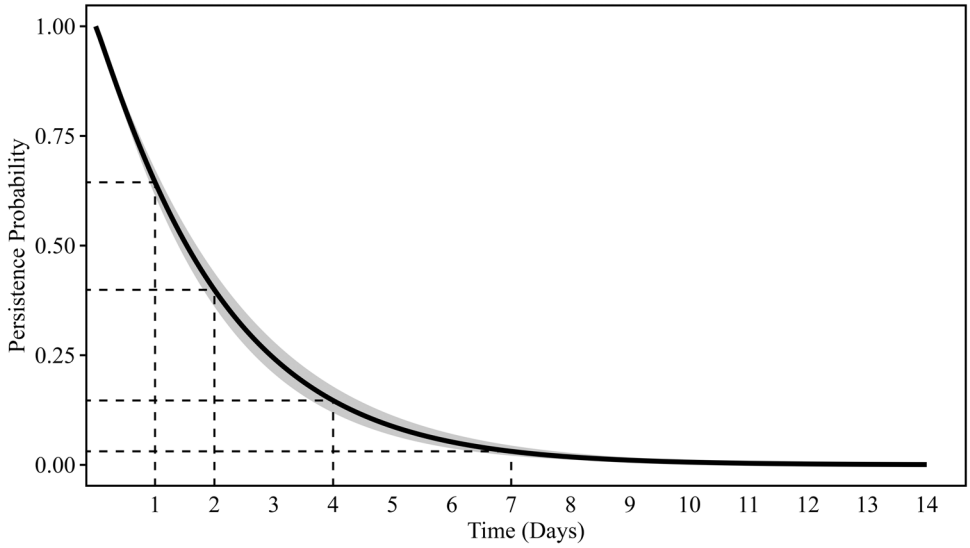


FIGURE 4 Predicted carcass persistence probability over monitoring time (in days) from the null Weibull survival model (solid black line) with 85% confidence interval (gray shading) for the 214 carcasses that were placed not in association with infrastructure on White Sands Missile Range in south-central New Mexico, USA, during spring (April–May) and fall (September–November) 2022. Dashed lines indicate persistence probability at 1, 2, 4, and 7 days following placement.

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a non-infrastructure-related scenario, similar to a recent regional mortality event in southern New Mexico. Our findings revealed that carcass persistence times were comparable to those reported near infrastructure (Barrientos et al. 2018), with >50% of carcasses scavenged within 2 days of placement. Additionally, none of the tested covariates were informative of carcass persistence times. Our findings suggest that persistence estimates from infrastructure-related studies may be generalizable to non-infrastructure-related contexts. Based on these results, we recommend that mortality surveys in non-infrastructure-related scenarios be conducted as frequently as possible (preferably every day), and that the data presented herein be used to adjust mortality estimates for scavenging losses.

While we expected carcass persistence times to be longer in our non-infrastructure-related study, we found them similar to those associated with infrastructure in previous studies (Costantini et al. 2017, Barrientos et al. 2018, Bernardino et al. 2022). For example, across 79 studies of infrastructure-related carcass persistence, Barrientos et al. (2018) found that approximately 70% of avian carcasses similar in size to those used in our study persisted after one day, while we found persistence probability to be 0.65 (85% CI = 0.59–0.70) for one day. However, as we did not measure persistence of carcasses associated with infrastructure in our study area, we cannot concretely confirm that persistence times between infrastructure-related mortality and non-infrastructure scenarios are similar. Future work in our study area should assess how carcass persistence varies for carcasses associated with and not associated with infrastructure to determine if persistence is similar between these 2 scenarios.

None of the variables we assessed were informative of carcass persistence times in our study. Although previous studies have reported seasonal variation in persistence times (Flint et al. 2010, Hager et al. 2012, Bispo et al. 2013, Borner et al. 2017), a meta-analysis by Barrientos et al. (2018) found that season had little or no influence, likely because a categorical variable of season does not adequately capture variability in scavenger activity or abundance. Additionally, previous studies on carcass persistence times have found varying effects of habitat (Barrientos et al. 2018). We predicted that carcass persistence times would increase with vegetation cover and visual obstruction, as denser vegetation could hinder carcass scavenging. However, none of the habitat variables we modeled (vegetation cover, visual obstruction, distance to road) were important predictors of carcass persistence times. The lack of relationship between habitat variables and persistence times may be explained by limited variability in vegetation structure and habitat across sites in our study system, and relatively low levels of human activity on roadways. Overall, vegetation cover and obstruction are relatively homogenous across our study sites and thus scavenger foraging abilities may not vary widely across this limited range of habitat structure. Human activity on roadways in our study system is relatively low, which may not produce enough roadkill to attract scavengers to road corridors, explaining the lack of relationship we found between persistence times and distance to roadways (Lambertucci et al. 2009, Santos et al. 2011, Fielding et al. 2021).

Photographic evidence of the scavenging event was not obtained for over half of the scavenged carcasses in our study, consistent with other studies (Paula et al. 2014, Rosa et al. 2019). However, our use of timelapse photos to document carcass removal still allowed for detection of a removal event within a short time interval, even when the exact scavenging event was not detected. Future studies should program camera traps to take photos at regular intervals (30 minutes) to maintain accuracy in quantifying removal time even when the exact removal event is not recorded. Carcasses without a photo of the scavenging event had shorter median persistence times than those with a photo, suggesting they may have been scavenged by avian or small mammal scavengers as carcasses scavenged by these guilds had the shortest median persistence times, or another guild, like reptiles, which were never captured on our cameras. The shorter persistence time for carcasses with no photo of the scavenging event, similar to the persistence time for carcasses scavenged by avian and small mammal scavengers, indicates that the camera traps employed in this study and others may not have been sensitive enough to trigger and photograph scavenging events by scavengers with smaller body sizes, possibly leading to underestimation of the importance of scavenger guilds in the carcass removal process.

It is important to note that the way in which carcasses were distributed on the landscape during our study does not represent all non-infrastructure related mortality scenarios, and our findings should be considered in this context. For our

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study, we modeled our placement of carcasses based on the mortality event that occurred in 2020 in our region, where carcasses were found at relatively low densities over a large area of the landscape over a period of multiple weeks. In some mass avian mortality scenarios caused by extreme weather, carcasses may appear on the landscape at high densities within small areas (Newton 2007), which would likely change the persistence time of those carcasses due to scavenger swamping. Additionally, we chose to use only one species of carcass in our study to match the primary size class of carcasses found during the 2020 mortality event. Carcass size has been determined to be an important predictor of persistence in other studies (Barrientos et al. 2018), and our findings may be less applicable for mortality events involving carcasses of different size classes. Lastly, due to access limitations for our study area, we placed all carcasses during daylight hours. In actual avian mortality scenarios, carcasses may appear at any time of the day or night. Scavenger activity and the species of scavenger active on the landscape varies between day and night, which may lead to variation in the persistence time of carcasses that are placed during daylight hours or at night. Future work should assess how carcass persistence in non-infrastructure related scenarios varies for carcasses of different sizes, for different densities of carcasses on the landscape, and should randomize carcass placement times throughout the day and night.

MANAGEMENT IMPLICATIONS

Camera traps are a useful tool for quantifying avian carcass persistence times, as evidenced by our study, but there are important factors to consider when implementing camera trap monitoring. First, for studies quantifying carcass persistence, camera traps that can be programmed to take both timelapse and motion-triggered photos should be used. The interval between timelapse photos should be as short as possible to minimize error for scavenging events that are not detected by the motion trigger, but long enough to ensure that data storage is not depleted too quickly. Motion-trigger sensitivity should be set to the most sensitive settings allowable for the surrounding vegetation structure to improve the odds of documenting all scavenging events as they occur.

Our study revealed that the persistence times of avian carcasses not associated with infrastructure is short, with over half of carcasses removed within 48 hr of placement, which has important implications for the design of mortality surveys. While correction factors can assist in more accurately estimating the number of carcasses present initially by correcting for scavenging loss (albeit imperfectly, see Ravache et al. 2024), important information, like avian species, is lost when carcasses are scavenged prior to surveys. In many instances, species-specific data from mortality surveys is critical to monitoring and management efforts. Therefore, it is preferable that surveys are conducted daily, and we recommend that future survey efforts are designed to survey sites as often as possible. Additionally, even if surveys are conducted daily, correcting mortality estimates for scavenging loss is critical for improving the accuracy of these estimates. The data provided herein allow for this correction, but future work must assess how persistence times vary between systems in non-infrastructure-related scenarios, and if persistence estimates are generalizable across systems.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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ETHICS STATEMENT

Animal capture and handling followed methods approved by New Mexico State University's Institutional Animal Care and Use Committee (protocol number 2022-028).

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